

50Days-coding-challenge

Day-37[Python]

Student Report System

Name:M NITHIN

Challenge:

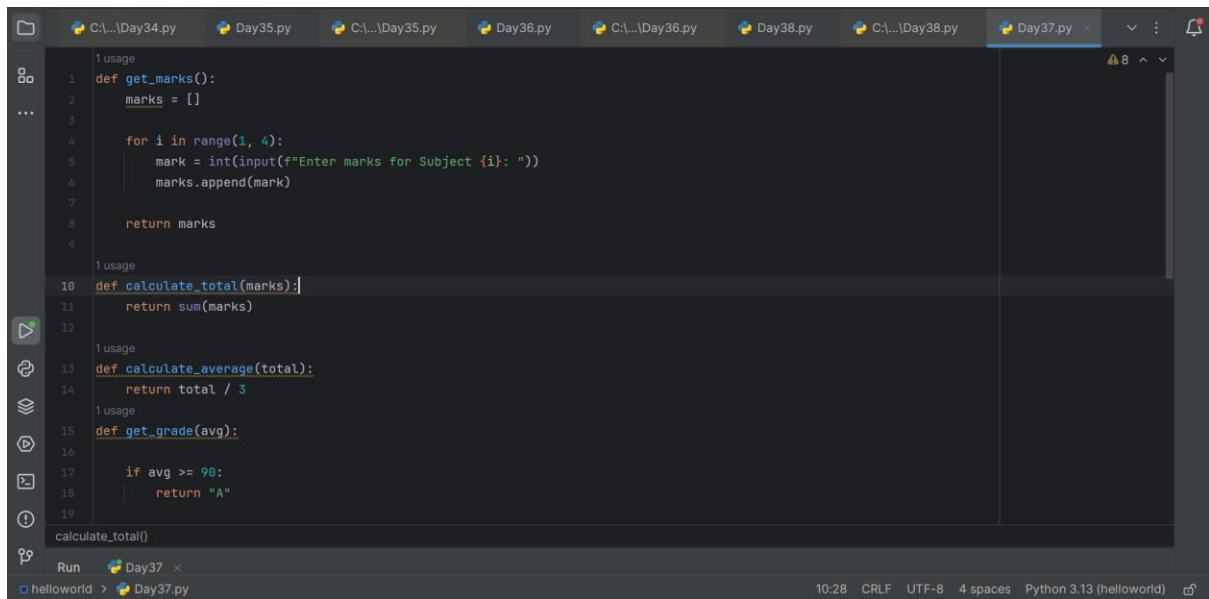
Write a Python program using user-defined functions to manage student marks.

Requirements:

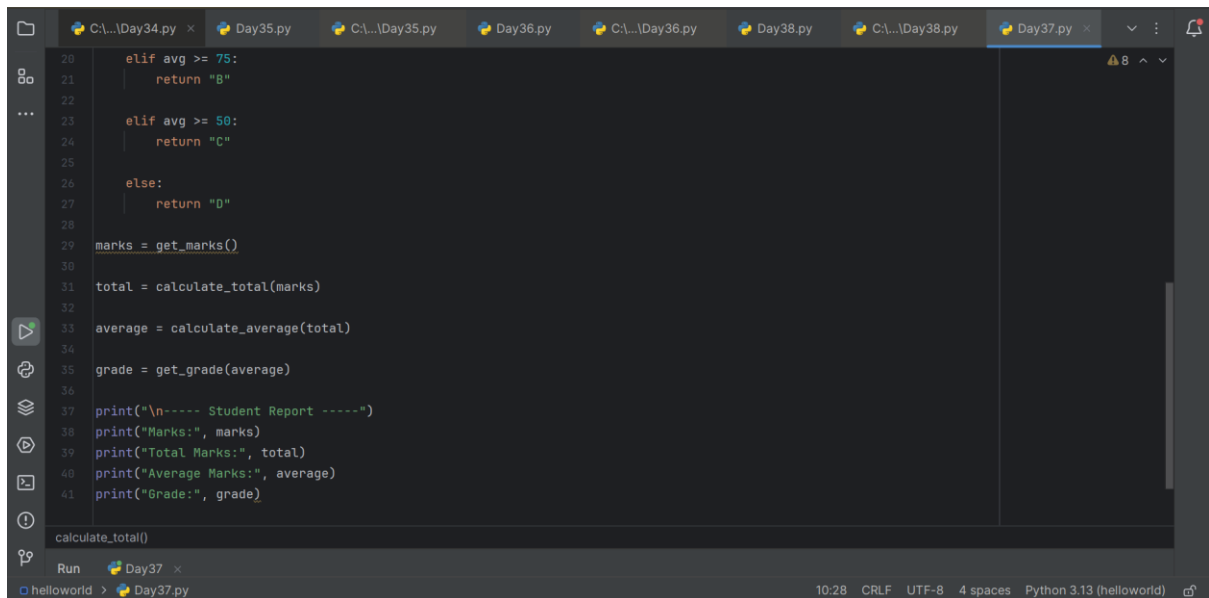
1. Create a function `get_marks()`
 - Takes input for marks of 3 subjects
 - Returns them as a list
2. Create a function `calculate_total(marks)`
 - Takes the list of marks
 - Returns the total
3. Create a function `calculate_average(total)`
 - Returns the average marks
4. Create a function `get_grade(avg)`
 - Returns grade based on:

$A \rightarrow \text{avg} \geq 90$
 $B \rightarrow \text{avg} \geq 75$
 $C \rightarrow \text{avg} \geq 50$
 $D \rightarrow \text{avg} < 50$
5. Main program should:
 - Call all functions
 - Display total, average, and grade

Results:



```
1 usage
2 def get_marks():
3     marks = []
4     for i in range(1, 4):
5         mark = int(input(f"Enter marks for Subject {i}: "))
6         marks.append(mark)
7
8     return marks
9
10 usage
11 def calculate_total(marks):
12     return sum(marks)
13
14 usage
15 def calculate_average(total):
16     return total / 3
17
18 usage
19 def get_grade(avg):
20     if avg >= 90:
21         return "A"
22
23 calculate_total()
24 Run Day37 x
25 helloworld > Day37.py
26 10:28 CRLF UTF-8 4 spaces Python 3.13 (helloworld)
```



```
20 elif avg >= 75:
21     return "B"
22
23 elif avg >= 50:
24     return "C"
25
26 else:
27     return "D"
28
29 marks = get_marks()
30
31 total = calculate_total(marks)
32
33 average = calculate_average(total)
34
35 grade = get_grade(average)
36
37 print("\n----- Student Report -----")
38 print("Marks:", marks)
39 print("Total Marks:", total)
40 print("Average Marks:", average)
41 print("Grade:", grade)
42
43 calculate_total()
44 Run Day37 x
45 helloworld > Day37.py
46 10:28 CRLF UTF-8 4 spaces Python 3.13 (helloworld)
```

Output:

The screenshot shows an IDE with a project named 'helloworld'. The main editor displays a Python script with the following code:

```
20     elif avg >= 75:
21         return "B"
22
23     elif avg >= 50:
24         return "C"
25
26     else:
27         return "D"

```

Below the editor, the 'Run' tab is active, showing the execution output for 'Day37.py'. The output is as follows:

```
C:\Users\NITHIN\PycharmProjects\helloworld\.venv\Scripts\python.exe C:\Users\NITHIN\PycharmProjects\helloworld\Day37.py
Enter marks for Subject 1: 88
Enter marks for Subject 2: 85
Enter marks for Subject 3: 99

----- Student Report -----
Marks: [88, 85, 99]
Total Marks: 272
Average Marks: 90.66666666666667
Grade: A

Process finished with exit code 0

```

The status bar at the bottom indicates the file is 'Day37.py' in the 'helloworld' project, using Python 3.13 with CRLF line endings, UTF-8 encoding, and 4 spaces for indentation.